



Society of Petroleum Engineers

SPE-230072-MS

Beyond Proprietary Systems: Realizing the Potential of Open Process Automation (OPA) for Next-generation Industrial Control

F. Al Kalbani, A. Al Abri, and N. Al Hamimi, Petroleum Development Oman (PDO), Muscat, Sultanate of Oman

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230072-MS](https://doi.org/10.2118/230072-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This paper presents a comprehensive overview of Open Process Automation (OPA), an industry-driven initiative aimed at transforming industrial control systems through the adoption of open architecture, standards-based approaches, and enhanced business practices. OPA seeks to address critical challenges in legacy control systems, such as lack of interoperability, high maintenance costs, and insufficient cybersecurity measures. The Open Process Automation Standard (O-PAS™), developed by The Open Group, provides a framework for scalable, modular, interchangeable, secure-by-design, and interoperable distributed control systems.

This work synthesizes insights from a field pilot initiative between Schneider Electric and Petroleum Development Oman (PDO). The pilot project specifically demonstrates O-PAS requirements, leveraging IEC61499 architecture to implement a multi-vendor architecture. By integrating these perspectives, this paper highlights the advancements and practical applications of OPA, emphasizing its potential to reduce total cost of ownership, increase operational profitability, and foster innovation in the industrial automation landscape.

Introduction

The landscape of process automation has historically been dominated by proprietary and closed control systems. While these legacy systems have proven reliable, their inherent closed nature presents significant challenges, including high upgrade and maintenance costs, limited interoperability between different vendors' systems, and an inability to reuse configurations or applications across varied product generations. Furthermore, cybersecurity, often an afterthought in the design of these systems, has become a critical concern in an increasingly interconnected industrial environment.

In response to these limitations, the Open Process Automation (OPA) initiative has emerged as a transformative force. OPA aims to redefine industrial control systems by establishing a standards-based, open, secure, and interoperable architecture. This initiative is driven by a consensus-based standards organization, The Open Process Automation Forum of The Open Group, which has been actively developing the Open Process Automation Standard (O-PAS™). The O-PAS standard seeks to enable the development of scalable, modular, interchangeable, and secure-by-design distributed control systems.

The adoption of OPA is motivated by the pressing need for industrial manufacturers to reduce capital and operational expenditures while simultaneously enhancing operational profitability. Open, interoperable systems foster increased competition among suppliers, leading to reduced costs and greater choice. They also facilitate the seamless integration of best-in-class, third-party components, thereby promoting innovation and flexibility. This paper synthesizes insights from recent advancements and field pilot initiatives, demonstrating how OPA, by embracing principles of heterogeneity, inherent security, multi-vendor interoperability, and future-proof innovation, offers a clear pathway for system migration and ultimately, greater value and profitability for end-users.

Open Process Automation Standard (OPA)

The Open Process Automation (OPA) initiative is fundamentally driven by a commitment to establish a standards-based, open, and secure architecture for industrial control systems. This approach is designed to overcome the limitations of traditional proprietary systems by fostering interoperability, modularity, and portability. At its core, the OPA standard defines a set of quality attributes crucial for next-generation automation systems. These include interoperability, scalability, affordability, availability, modularity, securability, portability, discoverability, standards conformance, and reliability. While all these attributes are vital, interoperability, modularity, and portability are particularly distinguishing features when compared to existing commercial distributed control systems (DCS) and programmable logic controllers (PLC).

The OPA standard adopts a "standard of standards" philosophy, meaning it integrates existing, proven industry standards wherever possible. New standards are only developed when no suitable existing standard is available. The Open Process Automation Forum actively collaborates with various organizations, including NAMUR, ZVEI, PLCopen, ISA, and OPC Foundation, to ensure broad alignment and leverage established expertise. Key industry standards incorporated into O-PAS include IEC 62264 (ISA 95) for technical architecture, IEC 62443 (ISA 99) for security, IEC 62541 (OPC UA) for the connectivity framework, DMTF (Redfish®) for system management, and IEC 62714 (AutomationML), IEC 62682 (ISA 18), IEC 61131, and IEC 61499 for information and exchange models. The physical platform, encompassing input/output modules and DCS/PLC controllers, is an area where applicable standards are still emerging, with efforts focused on defining attributes like size, weight, power, and connectors, drawing parallels from standards such as Sensor Open Standards Architecture (SOSA™) (Abbot, Bezenek, Borton, et al., 2021) and Peripheral Component Interconnect Express (PCI Express) (Budruk, Anderson, & Shankey, 2004).

Central to the OPA framework is its reference architecture as shown in [Figure 1](#), which introduces three novel elements compared to the traditional 3-level Purdue Reference Model: the Distributed Control Node (DCN), the O-PAS Connectivity Framework (OCF), and the Advanced Computing Platform (ACP). As per [Figure 2](#), the DCN is an O-PAS-conformant device responsible for input/output and user-programmable applications, connecting physically and data-wise to the OCF network. The OCF serves as the industry-standard based control network, while the ACP (sometimes referred to as Realtime Advanced Compute Platform or RTAC) provides the necessary computing power. The standard also defines various profiles and configurations for DCNs to accommodate diverse operational needs.

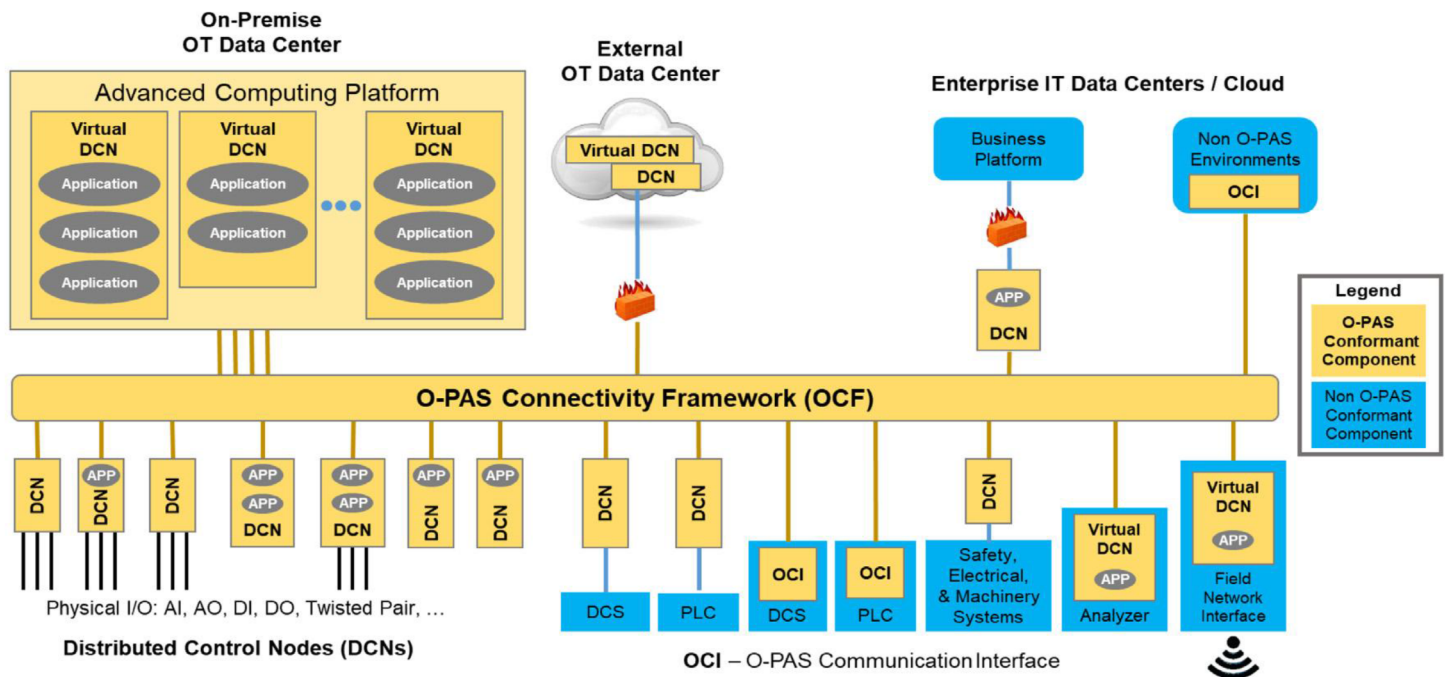


Figure 1—Open Process Automation reference architecture

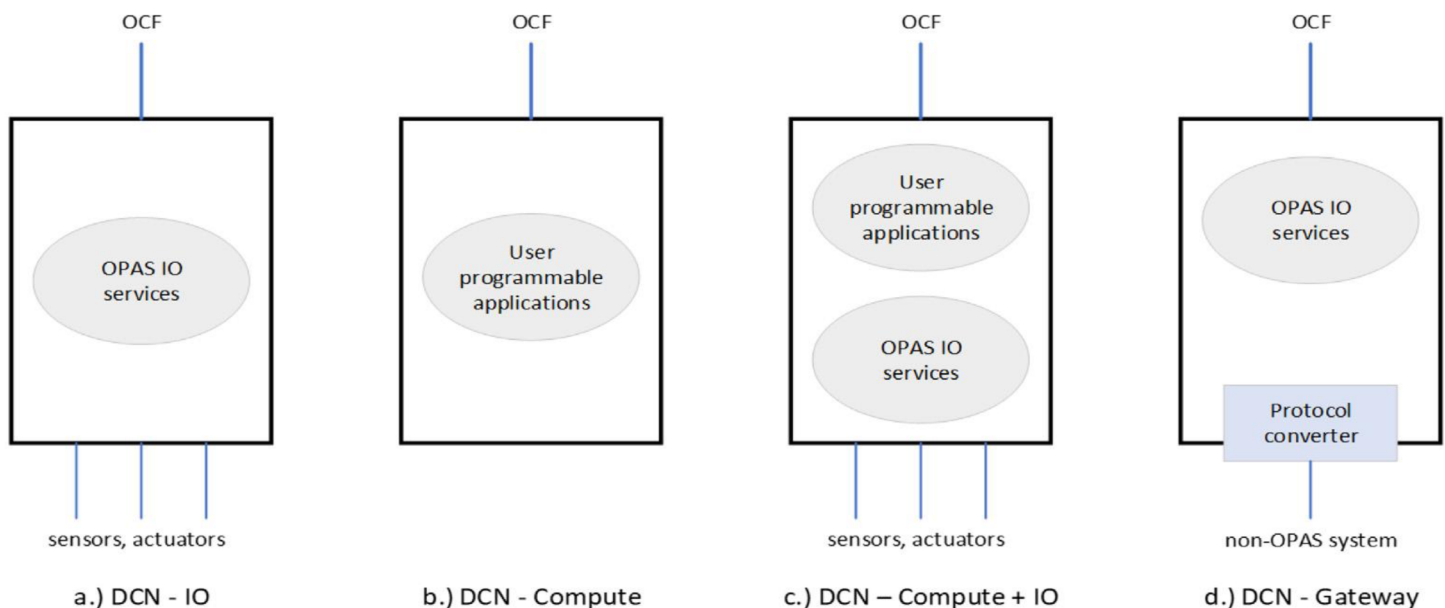


Figure 2—Distributed Control Node (DCN) profiles

Furthermore, the IEC 61499 architecture plays a significant role as a component solution for distributed industrial automation systems, promoting portability, reusability, interoperability, and reconfiguration of distributed applications. This standard offers a generic model for distributed systems, where applications are constructed from networks of Function Blocks. Complementing this, UniversalAutomation.org (UAO) is an independent, non-profit association dedicated to managing the implementation of an industrial automation shared-source runtime execution engine based on the IEC 61499 standard. This collaborative effort provides a foundation for an ecosystem of portable, interoperable, "plug and produce" solutions, thereby creating a new paradigm within industrial automation.

Methods, Procedures, and Processes

This section details the methodology and processes employed in the field pilot initiative, a collaborative effort between Schneider Electric and Petroleum Development Oman (PDO), aimed at demonstrating the practical application of the Open Process Automation Standard (O-PAS™). The pilot was strategically segmented into a series of Minimal Viable Products (MVPs) to systematically identify and address integration challenges as the O-PAS standard matures. The initial MVP (MVP-0) primarily focused on validating the core O-PAS requirements of interoperability and portability, with subsequent MVPs slated to explore other critical aspects such as security, system management, orchestration, and alarm handling.

Hiba Production Station Process

The pilot project was implemented at the Hibah Production Station, specifically focusing on Unit-00, which operates as a closed-loop system for water lifting. The primary objective of this unit is to extract water from the wellhead (W-0020) using compressed air, followed by the separation of the water/air mixture. The separated air is then vented to the atmosphere, while the processed water is recirculated back to the wellhead. This closed-loop operation relies on compressed air, supplied by reciprocating compressors, as the motive fluid to lift water and maintain optimal pressure within the plant equipment.

The operational flow involves the air/water mixture traveling from the wellhead to a production train. This train consists of a two-phase separator (V-0020), a surge tank (T-0020), and three shipping pumps (P-0020, P-0021, P-0022). A fixed wellhead, designed as a replica of a typical production wellhead, is integrated into the system. Water discharged from the shipping pumps flows into the bottom of the wellhead casing. Compressed process air is then introduced into the upper section of the wellhead, where it mixes with the water, thereby pressurizing the system. The resulting water/air mixture exits the wellhead through a wing valve and a hand-operated choke valve, which is used to regulate the process throughput.

Within the separator (V-0020), air is efficiently separated from the water and subsequently released to the atmosphere via a pressure control valve. The separated water then proceeds to the surge tank (T-0020) through a level control valve. The surge tank itself operates at atmospheric pressure, with its water level manually monitored and controlled by the activation or deactivation of the shipping pumps. These pumps can be initiated locally and controlled (stopped) either locally or remotely from the central control system. The water accumulated in the surge tank is then pumped back to the wellhead by the three centrifugal shipping pumps. A crucial element of this system is a recycle line located at the pumps' common discharge header, which ensures the pumps maintain their minimum flow requirement by returning water to the surge tank through a flow control valve.

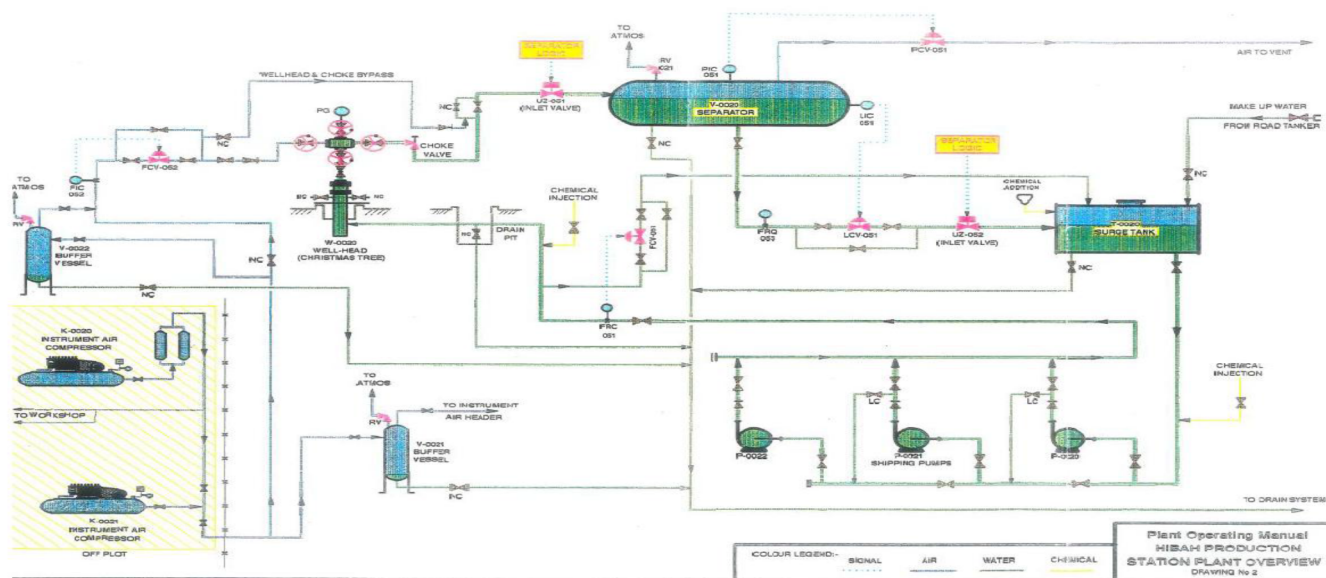


Figure 1 Hibah Production Station Control Overview Diagram

Figure 3—Control Overview Diagram

O-PAS Aligned System

The O-PAS aligned system for MVP-0 was specifically engineered to showcase the interoperability and portability benefits inherent in the O-PAS framework. A multi-vendor architecture was adopted, leveraging the Universal Automation Organization (UAO) IEC61499 run-time environment. This architecture incorporates various Distributed Control Nodes (DCNs), each representing a type of O-PAS conformant DCN. Figure 4 below shows the architecture

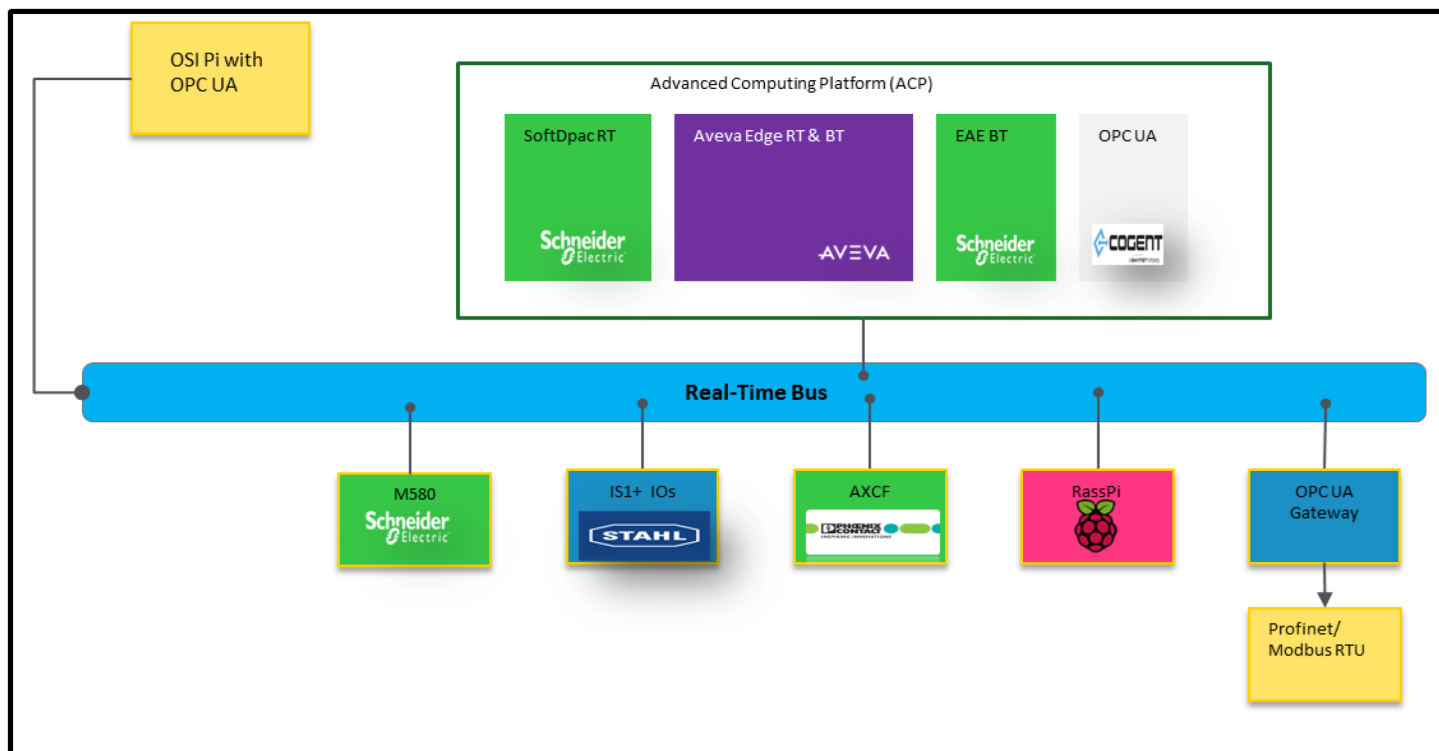


Figure 4a—Block diagram System Architecture

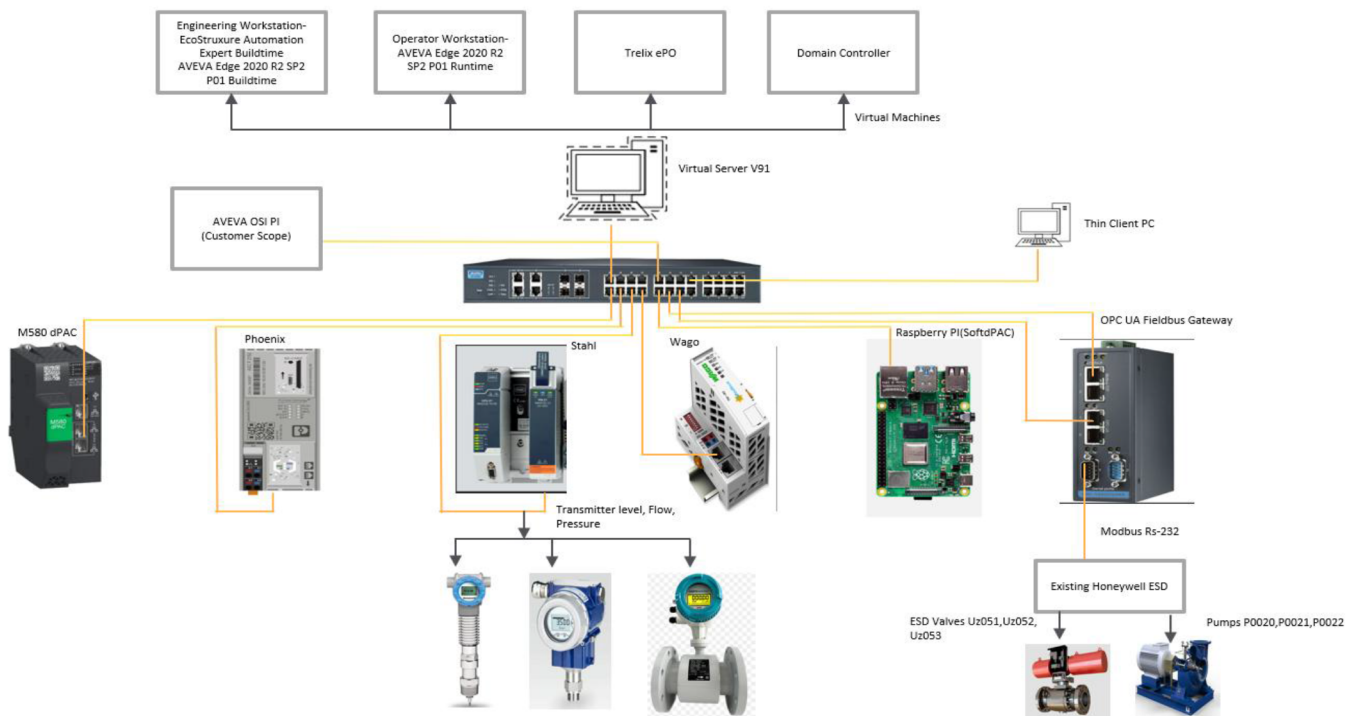


Figure 4b—System Architecture

The system architecture for the PDO O-PAS aligned system involves several components within Schneider Electric's scope. A virtualization server, V91, hosts virtual machines, while the engineering workstation manages control configuration using EcoStruxure Automation Expert and HMI development using AVEVA Edge 2020 R2 SP1. The operator workstation includes AVEVA Edge Runtime for the operator interface. Moreover, an Advantech Fieldbus gateway enables communication with the existing ESD system via Modbus Rs-232 protocol, converting it to an OPC UA signal for integration with our OPAS System. The architecture also encompasses various controllers such as M580dPAC, Stahl, Phoenix, Wago, and SoftdPAC, interconnected through the network and communicating via the IEC 61499 Standard. Analog transmitters and control valves are linked to the Stahl controller from the field. Other discrete signals are hardwired to the existing Honeywell system, providing device status information to the ESD system through communication. Lastly, the system receives and processes data for HMI, transmitting the information to AVEVA Edge through OPC UA communication. Additionally, there is a Thin Client station for the operator interface, which is connected to the network. All the field devices data are exposed through OPC UA to connect with the AVEVA OSI PI.

The plant is currently operated using the existing ESD system, which interfaces with the OPAS system through a Fieldbus Gateway. According to Figure 5, this gateway converts Modbus RS-232 signals into OPC UA protocol, enabling seamless communication with the OPAS system. Through this integration, the OPAS system can retrieve device status information from the ESD system and present it on the HMI for operator visibility. Furthermore, the HMI will also facilitate command transmission for the MOS signal and provide control over the pump's start/stop operations.

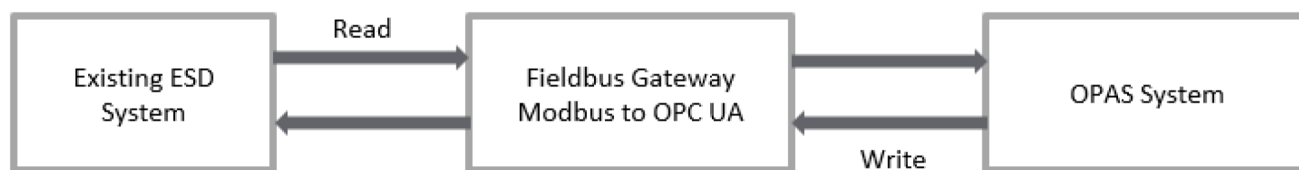


Figure 5—Third Party System Diagram

The selection of diverse DCNs was critical for demonstrating a true multi-vendor system and enabling the distribution of control applications across these nodes. The flexibility offered by remote I/O was instrumental in determining the optimal distribution of control applications, as these I/O units were accessible to all DCNs, including container-based soft controllers running on platforms like the Raspberry Pi. This level of flexibility and distribution was significantly facilitated by the adoption of IEC61499 as the core implementation standard. The primary communication protocol employed was OPC UA, facilitating communication between DCNs and HMIs, as well as between DCNs and remote I/O via an OPC UA gateway. Cross-communication between different DCNs was initially via UDP protocol, with plans for full OPC UA compliance in future O-PAS compliant components.

Interoperability was rigorously validated through multiple test scenarios. For instance, the control application was segmented into several sub-applications and distributed across different DCNs. The seamless exchange of data between these DCNs, regardless of whether they were from the same vendor, different vendors, or were software-based or embedded, confirmed the system's robust interoperability.

Control Application Development

The control application was developed using an asset modeling approach, leveraging the object-oriented programming capabilities within the EcoStruxure Automation Expert (EAE) build-time environment. This approach involved encapsulating all directly attached and controlling elements for a device within a dedicated function block, which solely describes that specific device. This methodology allowed for the creation of a comprehensive model of the entire control system, accurately representing the physical plant equipment. The configuration between these equipment-based function blocks was managed through data input and output for efficient data exchange. The control application provided a general overview representing various assets, such as the separator (V020), wellhead (W020), surge tank (T020), and shipping pumps (P020, P021, P022), encapsulated within Composite Automation Type (CAT) function blocks as shown in Figure 6.

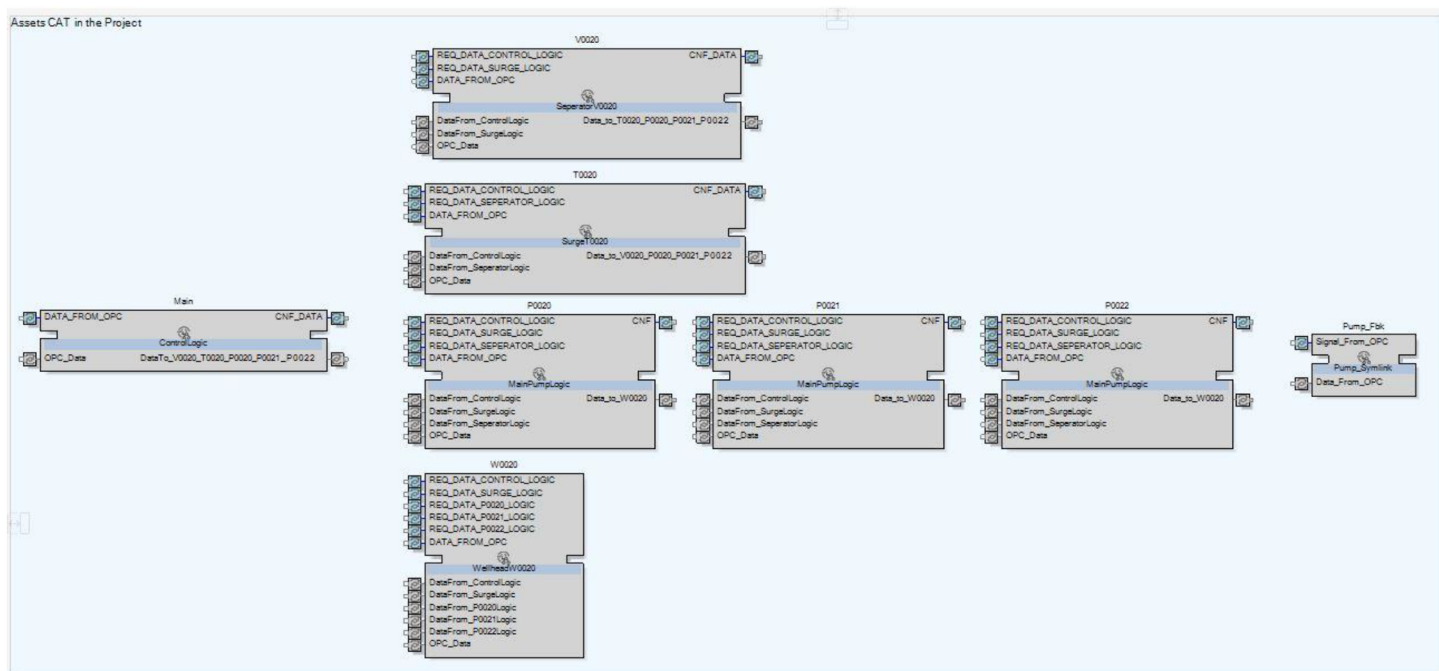


Figure 6—IEC61499 Control Application Overview

Process Model Simulation

A detailed process model was constructed to accurately mimic the behavior of the physical plant. This simulation model served multiple critical purposes: it facilitated the rigorous testing and validation of the control application prior to its deployment in the physical plant, aided in the fine-tuning of control loops, and provided an approximate assessment of the performance and parameters of various control strategies. Figure 7 below shows the process model for Hiba production plant

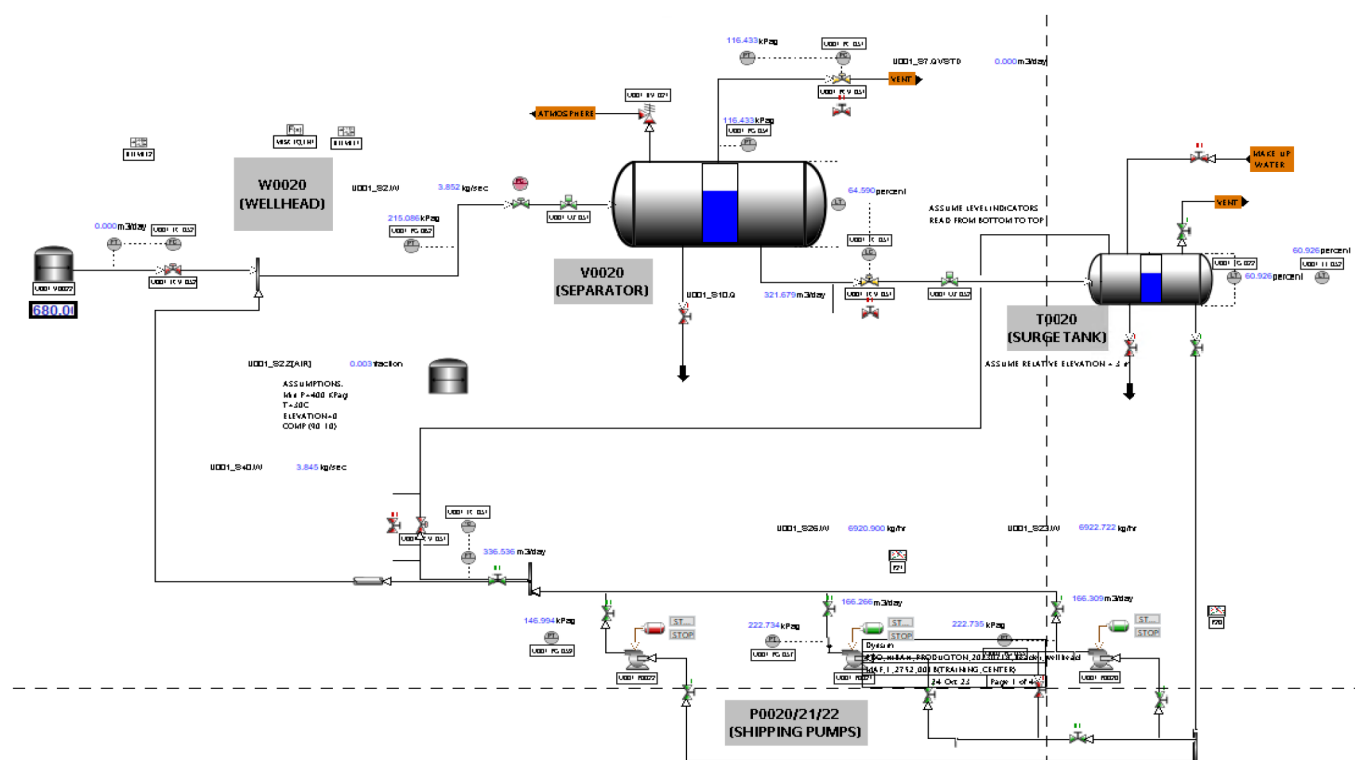


Figure 7—Process Model

Human Machine Interface

The Human Machine Interface (HMI) was developed using AVEVA Edge SCADA, serving as the primary interface for operators to manage and control the process. Communication between the HMI and the DCNs was established via OPC UA, ensuring reliable and standardized data exchange. Figure-8 below shows the overview of the HMI. The image below displays the process screen of the OPAS Hibah Training System, which shows the different instruments present in the process with their tag name and the status of those instruments. Discrete signals and Emergency Shutdown (ESD) valves are monitored only. For the pump, we can control the start/stop commands directly from the faceplate. Additionally, we have full control over PID, analog signals, and control valves through our HMI interface. The symbol animations adhere to the PDO guidelines.

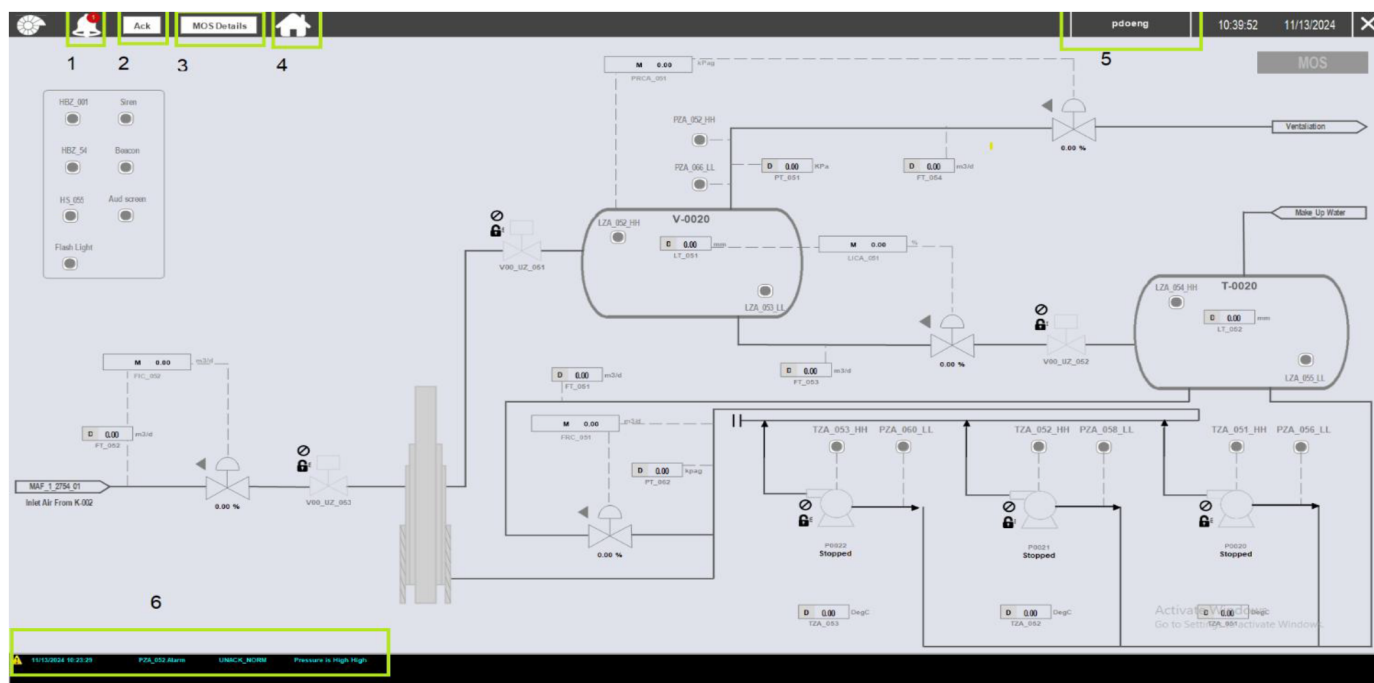


Figure 8—HMI Application Overview

In the above process screen overview taskbar, you can see the different icon with markup the details of the icons or below.

1. **Alarm Tab:** This screen displays the current alarms and their statuses, including details such as activation time, tag name, state, and message.
2. **Ack:** The Ack button is used to acknowledge all alarms.
3. **MOS Detail:** This screen shows various MOS signals and their statuses, indicating whether they are enabled or not.
4. **Home:** This option takes you back to the home process screen.
5. **Login:** This section displays the currently logged-in user. You can also change the user through this option.
6. **Alarm Tab:** This feature allows you to view current alarms without needing to open the Alarm screen.

MOS screen overview

The MOS Detail screen provides information about MOS signals, including their alarm status, MOS command status, and overall, MOS status.



Figure 10—Alarm Overview

In the above Alarm Overview screen, you can see the different markup their details are below.

1. The alarm animation shows that the alarm was triggered and acknowledged.
2. The alarm animation indicates that the alarm was triggered but has not been acknowledged.
3. The alarm animation signifies that the alarm was triggered.

Conclusion

The Open Process Automation (OPA) initiative is a significant driver of innovation, largely by integrating approaches, technologies, and standards from adjacent industries and cutting-edge academic research. This cross-pollination of ideas and technologies is crucial for developing a truly open and future-proof industrial control system. Notable inputs include the adoption of Modular Open Systems Architectures, exemplified by the Future Airborne Capability Environment (FACE™) standard from the avionics industry, and Network Function Virtualization, a concept widely used in telecommunications. These adoptions highlight OPA's commitment to leveraging established, robust frameworks from other complex domains.

Furthermore, the O-PAS standard incorporates enabling technologies derived from recent academic and industrial research. These include advancements in deterministic, real-time networking, the IEC 61499 standard for distributed industrial automation, and the OPC UA (Open Platform Communications Unified Architecture) for secure and reliable data exchange. The objective is to provide the necessary computing platforms and a supportive commercial ecosystem that will facilitate a step-change improvement in the timeliness and extent of industrial adoption of advanced control strategies, such as Model Predictive Control (MPC), and the next generation of machine learning-based process system engineering technology.

The field pilot initiative, a collaborative effort between Schneider Electric and Petroleum Development Oman (PDO), serves as a concrete demonstration of these innovative aspects. This case study specifically aimed to showcase key requirements of the O-PAS™ Standard, positioning it as an emerging next-generation process automation system. By redefining the architecture of industrial automation and control systems, the O-PAS™ Standard enables the development of systems that are inherently scalable, modular, interchangeable, secure by design, and interoperable. The pilot's success in leveraging the IEC61499

architecture and the UniversalAutomation.org (UAO) initiative to achieve its objectives and implement a multi-vendor architecture further validates the innovative potential and practical applicability of the OPA framework.

One of the key limitations found of the OPAS system in terms of cybersecurity is its reliance on standardized, open communication protocols like OPC UA, which, while enabling interoperability, can also increase exposure to cyber threats if not properly secured. Additionally, integration with legacy systems through gateways (e.g., Modbus RS-232) may introduce vulnerabilities, especially if those legacy systems lack robust authentication or encryption. Without stringent network segmentation, access controls, and regular security updates, the OPAS system could become a potential entry point for cyberattacks within the OT environment.

Acknowledgment

The work described in this paper was carried out as a part of a lean project, under the sponsorship of engineering function department in PDO Oman and Ministry of Energy and Minerals (MEM) .

Reference

1. Abbot, D., Bezenek, G., Borton, N., et al (2021). *Technical standard for SOSATM reference architecture*. Version 1.0. The Open Group. <https://publications.opengroup.org/c212>.
2. Budruk, R., Anderson, D., & Shankey, T. (2004). *PCI express system architecture*. Addison-Wesley.